RCOTrackV1

Research Center For Oceanology (RCO) Lagrangian Trajectory Model for Marine Debris & Microplastic Simulation

USER MANUAL & COMPUTATIONAL TECHNICAL GUIDE

OFFICIAL CODE REPOSITORY:
https://github.com/dwygngrh/RCOTrack

MANDATORY OPERATING SYSTEM REQUIREMENT:
THIS PROGRAM MUST BE RUN IN A LINUX ENVIRONMENT (NATIVE LINUX / HPC SLURM / WSL2)

A 3D Lagrangian Particle Tracking Computational Engine Powered by JIT-Accelerated Python-Numba
With Assimilation of CMEMS Ocean Currents, ERA5 Winds, Biofouling Dynamics, and Beaching Processes
Original Code Created & Authored by Dr. Dwiyoga Nugroho

PROJECT RESEARCH INVESTIGATORS:

Research Center for Oceanography (RCO) – BRIN:

1. Dwiyoga Nugroho (Lead Model Developer & Principal Investigator)
2. Mochamad Riza Iskandar
3. Yoke Hany Restiangsih
4. Dede Falahudin
5. Agus Dendi Rochendi

Research Center for Deep Sea (RCDS) – BRIN:

1. Intan Suci Nurhati
2. Corry Yanti Manullang

Original code made by Dr. Dwiyoga Nugroho.
Acknowledgment: Assistance of Gemini AI (Google) was utilized for Numba Just-In-Time (JIT) compilation with multi-threaded CPU parallelization and vectorization, enabling real-time calculation of millions of particle steps without Python interpreter overhead.

The National Research and Innovation Agency (BRIN)
Jakarta, Indonesia —May 2026

TABLE OF CONTENTS OVERVIEW

Chapter 1: Introduction, Objectives & Research Investigators:

• Scientific background, computational scope, institutional research investigators, code authorship & specific AI assistance acknowledgment, and modular software architecture.

Chapter 2: Source Code Acquisition & System Installation:

• Downloading the code from GitHub (https://github.com/dwygngrh/RCOTrack), MANDATORY Linux OS requirement, Conda setup, and dependency installation.

Chapter 3: Dimensionality Frameworks (2.5D, 3DUV, 3DUVW):

• In-depth physics, equations, hydrodynamic requirements, and selection guide for 2.5D, 3DUV, and 3DUVW dimensionality modes.

Chapter 4: Simulation Configuration & Parameter Setup (Namelist):

• Structure of namelist.txt, global simulation parameters, forward vs. backward tracking, and physical diffusion/release settings.

Chapter 5: Input Dataset Specifications:

• CMEMS hydrodynamic 3D velocity fields, ERA5 10m surface winds, and river/marine particle seeding CSV files.

Chapter 6: Mathematical Formulations & Particle Dynamics:

• Equations: Lagrangian kinematic equations, numerical schemes (Euler, RK2, RK4), wind leeway drift, turbulent diffusion, settling velocity, biofouling, and beaching.

Chapter 7: Simulation Execution & Output Management:

• Operational workflow for executing main.py in Linux Bash, console progress tracking, memory management (garbage collection), and NetCDF-4 CF-compliant output structure.

Chapter 8: Post-Processing & Spatial Visualization:

• Plotting marine debris trajectories (plot_marine_debris.py), 3D microplastic visualization with inverted colorbar (plot_microplastic.py), and GIF animation generation.

Chapter 9: Troubleshooting & Technical References:

• Common errors and debugging solutions, Linux pathing and permission checks, CFL stability criteria (OMML), offline GIS mapping configurations, and key scientific literature.

CHAPTER 1: INTRODUCTION, OBJECTIVES & RESEARCH INVESTIGATORS

1.1 Background and Scientific Objectives

Marine debris pollution and microplastic contamination across Indonesian waters and the broader Indo-Pacific region represent an urgent ecological and oceanographic challenge requiring robust quantitative transport modeling. The Indonesian maritime continent features one of the most complex archipelagic land-sea configurations in the world, characterized by narrow straits, energetic tidal regimes, the major pathways of the Indonesian Throughflow (ITF), and intense seasonal reversals driven by the Asian-Australian Monsoon. Quantifying particle dispersion, cross-basin trajectories, and ultimate stranding locations requires high-performance computational numerical tools.

The Research center For Oceanology Lagrangian Trajectory Model V1 (RCOTrackV1) was conceived, as an accurate, modular, and high-performance particle tracking engine. Engineered and optimized specifically for Linux High-Performance Computing (HPC) clusters and Linux scientific workstations, LTM supports forward tracking to predict dispersion from coastal and riverine sources, as well as backward tracking (adjoint trajectory tracking) to identify the upstream origins of debris intercepted in marine protected areas, coral reefs, or coastal infrastructure.

1.2 Project Research Investigators & Authorship Attribution

The development, hydrodynamic validation, observational calibration, and biogeochemical parameterization of the RCOTrackV1 (Research center For Oceanology Particle Tracking V1) model are conducted by a collaborative research team in the National Research and Innovation Agency (BRIN), Indonesia:

Investigator Name	Institutional Affiliation	Research Specialization / Role
Dr. Dwiyoga Nugroho	Research Center for Oceanography (RCO - BRIN)	Original Code Author, Principal Investigator & Lead Numerical Model Developer (Physical Oceanography / Lagrangian Dynamics)
Mochamad Riza Iskandar	Research Center for Oceanography (RCO - BRIN)	Co-Investigator (Ocean Dynamics & Coastal Hydrodynamic Modeling)
Yoke Hany Restiangsih	Research Center for Oceanography (RCO - BRIN)	Co-Investigator (Marine Environmental Assessment & Oceanographic Data Analysis)
Dede Falahudin	Research Center for Oceanography (RCO - BRIN)	Co-Investigator (Marine Ecotoxicology & Plastic Pollution Geochemistry)
Agus Dendi Rochendi	Research Center for Oceanography (RCO - BRIN)	Co-Investigator (Oceanographic Instrumentation, In-situ Sensing & Field Validation)

Dr. Intan Suci Nurhati	Research Center for Deep Sea (RCDS - BRIN)	Senior Research Investigator (Paleoceanography, Marine Climate & Environmental Geochemistry)
Corry Yanti Manullang	Research Center for Deep Sea (RCDS - BRIN)	Co-Investigator (Marine Plastic Pollution)

CODE AUTHORSHIP & AI ASSISTANCE ATTRIBUTION

Original code was created and authored by Dr. Dwiyoga Nugroho (RCO - BRIN). Acknowledgment with the help of Gemini AI (Google) was utilized for: Numba Just-In-Time (JIT) compilation with multi-threaded CPU parallelization and vectorization, enabling real-time calculation of millions of particle steps without Python interpreter overhead.

1.3 Scope and Core Capabilities

RCOTrackV1 incorporates state-of-the-art computational features and oceanographic physical parameterizations:

- **Multi-Dimensional Modeling Paradigms:** Offers three distinct dimensionality representations (2.5D, 3DUV, and 3DUVW) tailored to floating macro-debris, settling microplastics without explicit vertical currents, and full 3D hydrodynamic upwelling/downwelling simulations.
- **Accelerated High-Performance Computing:** Powered by Numba Just-In-Time (JIT) compilation with multi-threaded CPU parallelization and vectorization (developed with assistance from Gemini AI), enabling real-time calculation of millions of particle steps without Python interpreter overhead.
- **Flexible Numerical Integration:** Provides 1st-order Euler, 2nd-order Runge-Kutta (RK2 / Heun), and 4th-order Runge-Kutta (RK4) integration schemes to maintain trajectory precision across mesoscale eddies and narrow straits.
- **Microplastic Physics & Biofouling:** Incorporates settling dynamics (Kooi et al. model) based on standard polymer densities (PET, PE, PP, PS, PVC, PA), algal biofilm growth rates, and microbial biofilm densities.
- **Coastal Beaching & Stranding:** Evaluates stochastic stranding probability based on proximity to coastlines and local bathymetry depth to halt active advection for beached debris.
- **Standard NetCDF-4 Output:** Output data is formatted according to Climate and Forecast (CF) Metadata Conventions using NetCDF-4/HDF5, fully compatible with CDO, NCO, Xarray, Panoply, and QGIS.

1.4 Modular Software Architecture

The RCOTrackv1 software framework is organized into decoupled, modular components within the root repository:

- **main.py:** Controls the main temporal simulation loop, periodic particle seeding, execution of compiled hydrodynamic update kernels, memory garbage collection, and interactive tqdm progress logging.
- **Namelist/config_parser.py:** Parses and validates the text-based namelist configuration files, handling simulation dates, directory paths, physical constants, and seeding source locations.
- **IO/nc_handler.py:** Manages CMEMS hydrodynamic velocity fields and ERA5 wind datasets via xarray/netcdf4, performing spatial domain cropping, vertical layer indexing, and temporal interpolations.
- **IO/output_writer.py:** Initializes the output NetCDF-4 trajectory file, defines multidimensional variables (time, particle ID, status), and handles periodic step-by-step writes.
- **Initial/seeding.py:** Injects new particles into the computational domain according to CSV source coordinates, stochastic release rates, and active station time windows.
- **Hydrodynamic/kernels.py:** High-performance Numba JIT-compiled kernels executing 3D advection, bilinear velocity interpolation, wind leeway drift, Kh/Kz random walk diffusion, biofilm growth, and shoreline beaching tests.
- **Model/microplastic.py:** Provides standard polymer densities and dynamic settling velocity formulations (ws) for various microplastic geometries.

CHAPTER 2: SOURCE CODE ACQUISITION & SYSTEM INSTALLATION

2.1 Downloading the Program from GitHub

The complete source code, configuration templates, utility scripts, and input datasets for RCOTrack / LTM are officially hosted on GitHub and freely available for research and operational use.

> **OFFICIAL GITHUB REPOSITORY URL**
> Repository: https://github.com/dwygngrh/RCOTrack
> Original Author: Dr. Dwiyoga Nugroho (dwygngrh)
> License: Open Source Research License (Apache-2.0)

To download and clone the project onto your local Linux system or HPC cluster, run:

```
# Clone the RCOTrack repository using Git
git clone https://github.com/dwygngrh/RCOTrack.git

# Enter the cloned project directory
cd RCOTrack

# Alternatively, download and extract the ZIP archive via curl / wget
wget https://github.com/dwygngrh/RCOTrack/archive/refs/heads/main.zip -O
RCOTrack.zip
unzip RCOTrack.zip
cd RCOTrack-main
```

> **MANDATORY OPERATING SYSTEM: LINUX REQUIRED**
>
> RCOTrackV1 MUST be executed in a Linux environment. The model codebase, automated deployment scripts (install_library.sh), file path separators, POSIX shared-memory allocation, NetCDF-4/HDF5 C-bindings, and Numba JIT OpenMP thread-pool bindings are built and optimized exclusively for Linux POSIX systems. Running the program natively under Windows Command Prompt or Windows PowerShell is NOT supported. If working on a Windows workstation, you MUST run the program inside WSL2 (Windows Subsystem for Linux) running Ubuntu 20.04/22.04 LTS or inside a Linux Docker container.

2.2 Supported Linux Distributions & Hardware Sizing

The model is certified and supported on the following Linux environments:

- • **Primary Distribution:** Ubuntu Linux 20.04 LTS, 22.04 LTS, 24.04 LTS (Recommended for workstations and standalone compute nodes).
- • **Enterprise / HPC Clusters:** Rocky Linux 8/9, AlmaLinux 8/9, Red Hat Enterprise Linux (RHEL) 8/9, CentOS 7/8 (Standard for HPC Slurm clusters).
- • **Windows Subsystem for Linux:** Windows 10/11 running WSL2 (Ubuntu 22.04 distro). In WSL2, access your project directory via Linux paths (e.g., /mnt/d/RCOTrack/ or /home/user/RCOTrack/).

Recommended hardware sizing for regional and national oceanographic simulations:

- **Processor (CPU):** Minimum 4 cores; 8 to 32 cores recommended (Intel Xeon, AMD EPYC, or modern multi-core x86_64 processors) with AVX2 instruction set support.
- **System Memory (RAM):** Minimum 8 GB RAM for localized bays; 32 GB to 64 GB RAM recommended for full Indonesian domain simulations with high-resolution CMEMS (0.083°) and ERA5 (0.25°) grids.
- **Storage:** High-speed storage (NVMe SSD or HPC parallel scratch filesystem) with at least 50 GB free space for hydrodynamic NetCDF files and particle trajectories.

2.3 Automated Installation via Conda Bash Script

The repository includes an automated bash setup script named install_library.sh. This script creates an isolated Conda environment named 'lagrangian' running Python 3.10 and installs all required scientific, numeric, and oceanographic libraries from the conda-forge channel.

Execute the installation in the Linux Bash terminal:

```
# Grant execution permissions to the script
chmod +x install_library.sh

# Run the automated installation script in Linux Bash
./install_library.sh

# Activate the newly created conda environment
conda activate lagrangian
```

2.4 Inspection of install_library.sh

The configuration directives executed within install_library.sh are detailed below:

```
#!/bin/bash
# Create the conda environment with Python 3.10
conda create -n lagrangian python=3.10 -y

# Activate the environment
source activate lagrangian

# Install core data processing and oceanographic libraries
# xarray and netcdf4 are required for CMEMS and ERA5 NetCDF handling
# pandas handles temporal indexing for daily/hourly data
conda install -c conda-forge xarray netcdf4 pandas dask -y

# Install high-performance computing libraries
# Numba is required for parallelized JIT-compiled kernels
# Numpy handles coordinate and array vector math
conda install -c conda-forge numpy numba -y

# Install configuration and geospatial visualization utilities
conda install -c conda-forge configparser matplotlib cartopy pillow tqdm -y
```

2.5 Installation Verification in Linux

Verify that all critical dependencies are correctly installed in your Linux environment before initiating simulations by running:

```
python -c "import xarray, netCDF4, pandas, numpy, numba, cartopy; print('LTM
Computational Environment on Linux: READY!')"
```

CHAPTER 3: DIMENSIONALITY FRAMEWORKS: 2.5D, 3DUV, AND 3DUVW

RCOTrackV1 provides three distinct computational dimensionality frameworks via the 'dimension' parameter in namelist.txt: 2.5D, 3DUV, and 3DUVW. In the codebase, this parameter is mapped to an internal integer flag (dim_mode = 0 for 2.5D, 1 for 3DUV, and 2 for 3DUVW), directing kernel execution in update_3d_safe. Selecting the appropriate dimensionality is critical for physical realism, data availability, and computational efficiency.

Feature / Aspect	2.5D Mode (dim_mode = 0)	3DUV Mode (dim_mode = 1)	3DUVW Mode (dim_mode = 2)
Primary Application	Positively buoyant floating macro-debris, styrofoam, surface oil, wooden drifters, search and rescue.	3D microplastics, suspended particles, sedimenting matter in standard ocean reanalyses without vertical velocity.	Full 3D realistic ocean dynamics, deep convective mixing, upwelling/downwelling zones, estuarine circulation.
Vertical Coordinate (z)	Fixed at surface (z = 0 m). No vertical displacement.	Dynamic 3D depth z(t) updated at each step.	Dynamic 3D depth z(t) updated at each step.
Horizontal Currents (u, v)	Surface layer ocean currents: u(x, y, 0), v(x, y, 0).	3D velocity interpolated at particle depth: u(x, y, z), v(x, y, z).	3D velocity interpolated at particle depth: u(x, y, z), v(x, y, z).
Direct Wind Leeway Drag	Active: Lw · U10 and Lw · V10 directly added to surface velocity.	Inactive: Wind drag is 0; wind influences particles only indirectly via ocean currents.	Inactive: Wind drag is 0; wind influences particles only indirectly via ocean currents.
Vertical Ocean Velocity (w)	None (w_ocean = 0).	Omitted / Diagnostic (w_ocean = 0).	Active: Explicit vertical hydrodynamic current w(x, y, z) interpolated from model.
Settling / Sinking Velocity (ws)	Inactive (ws = 0).	Active: Stokes settling velocity (static or dynamic with biofouling).	Active: Stokes settling velocity (static or dynamic with biofouling).
Vertical Diffusion (Kz)	Inactive (Kz = 0).	Active: Random walk vertical turbulent diffusion (Kz > 0).	Active: Random walk vertical turbulent diffusion (Kz > 0).
Hydrodynamic Data Required	Surface uo, vo NetCDF + 10m u10, v10 NetCDF.	3D uo, vo (multi-depth levels) NetCDF.	3D uo, vo, wo (multi-depth levels with vertical velocity) NetCDF.

3.2 The 2.5D Surface Leeway Framework (dim_mode = 0)

The 2.5D formulation is designed specifically for floating marine debris whose positive buoyancy pins them to the air-sea interface. In this mode, particles are restricted to the 2D surface layer (z = 0), but the dynamics are termed '2.5D' because they integrate both 2D surface ocean currents and atmospheric wind drag vectors at 10 meters height:

$$\frac{dx}{dt} = u_{ocean}(x, y, 0, t) + L_w U_{10}(x, y, t) + U_{diff,x}$$

$$\frac{dy}{dt} = v_{ocean}(x, y, 0, t) + L_w V_{10}(x, y, t) + U_{diff,y}, \quad \frac{dz}{dt} = 0$$

Equation 3.1: 2.5D Surface Drift Kinematics with Wind Leeway

Key characteristics of 2.5D:

• The vertical position remains locked at the surface (z = 0 m). Vertical velocity w and settling velocity ws are forced to 0.

• Vertical turbulent diffusion Kz is disabled (kz = 0.0).

• Ideal for tracking macroplastics, bottles, buoys, floating fishing nets, and styrofoam fragments.

• Does not require 3D subsurface hydrodynamic data, significantly reducing data storage and I/O overhead.

3.3 The 3DUV Settling Framework (dim_mode = 1)

The 3DUV framework models full 3D spatial dispersion without requiring explicit vertical ocean velocity fields (wo). This is the most widely used configuration for microplastic research driven by operational global ocean reanalysis products (such as CMEMS GLORYS12V1 or Mercator Ocean), where horizontal velocity components (uo, vo) are well-constrained by data assimilation while vertical velocity (wo) is either omitted from public distributions, noisy, or excessively smoothed.

$$\frac{dx}{dt} = u_{ocean}(x, y, z, t) + U_{diff,x}, \quad \frac{dy}{dt} = v_{ocean}(x, y, z, t) + U_{diff,y}$$

$$\frac{dz}{dt} = w_s(t) + R_{n,z}\sqrt{2 \cdot K_z \cdot |\Delta t|}$$

Equation 3.2: 3DUV Kinematics with Terminal Settling and Vertical Diffusion

Key characteristics of 3DUV:

• Particles move horizontally at their instantaneous depth z, sampling depth-dependent current shear (e.g., surface Ekman currents vs. deep subsurface ITF flow).

• Vertical displacement is driven strictly by particle settling velocity ws (gravitational sinking/buoyant rising) and vertical turbulent diffusion Kz.

• Direct wind leeway is 0 (leeway = 0.0), as particles below the surface boundary layer do not experience direct atmospheric wind force.

• Highly practical and robust for microplastics in regional ocean models where 3D horizontal currents are accessible.

3.4 The 3DUVW Full Hydrodynamic Framework (dim_mode = 2)

The 3DUVW framework represents the most physically complete 3D Lagrangian formulation. In addition to 3D horizontal velocity (u, v), 3DUVW assimilates the vertical hydrodynamic velocity field w(x, y, z, t) directly from numerical ocean circulation models (such as CROCO, ROMS, NEMO, or specialized CMEMS products with vertical velocity output):

$$\frac{d\mathbf{r}}{dt} = \mathbf{u}(x,y,z,t) + w_s\mathbf{k} + \mathbf{U}_{diff}$$

$$\frac{dz}{dt} = w_{ocean}(x,y,z,t) + w_s(t) + R_{n,z}\sqrt{2 \cdot K_z \cdot |\Delta t|}$$

Equation 3.3: 3DUVW Full Hydrodynamic Vertical Motion Equation

Key characteristics of 3DUVW:

• Captures active oceanographic vertical transport mechanisms, including coastal upwelling (e.g., South Java Upwelling during the Southeast Monsoon), downwelling frontogenesis, internal wave pumping, and deep convective overturning.

• Negatively buoyant microplastics can be resuspended into upper layers by strong upwelling currents exceeding their settling velocity (|w_ocean| > |ws|).

• Positively buoyant particles can be subducted below the euphotic zone in intense convergence and downwelling zones.

• Requires input NetCDF files containing the vertical velocity variable 'wo'.

CHAPTER 4: SIMULATION CONFIGURATION & PARAMETER SETUP (NAMELIST)

4.1 Namelist File Structure

All operational parameters for RCOTrackV1 are managed through plain-text namelist configuration files (.txt) structured in standard INI format. Users can switch simulation modes, adjust time steps, modify particle types, and choose integration schemes without editing Python source code.

Summary of namelist sections and their operational roles:

- **[Startup]:** Sets model branding and log headers.
- **[Simulation]:** Configures project name, simulation mode, dimensionality (2.5D, 3DUV, 3DUVW), time step dt, numerical scheme, and date boundaries.
- **[Diffusion]:** Controls horizontal (Kh) and vertical (Kz) turbulent diffusion coefficients.
- **[Files]:** Specifies Linux filesystem paths to CMEMS data, ERA5 data, output directory, and source CSV files.
- **[Release]:** Controls release dates, daily seeding rates, and particle release diameters.
- **[MarineDebris]:** Active in 'marine_debris' mode: debris type, leeway factor, and beaching probability.
- **[Microplastic]:** Active in 'microplastic' mode: polymer type, shape, settling velocity, and biofouling parameters.

4.2 Example Configuration: Floating Marine Debris (2.5D Forward RK4)

The configuration file namelist_marine_debris_forward_RK4.txt demonstrates a forward macroplastic simulation:

```
[Startup]
branding = Lagrangian Trajectory Model, RCO-BRIN, Indonesia-2026

[Simulation]
project_name = test_run
mode = marine_debris
dimension = 2.5D
dt = 3600              # Positive dt indicates FORWARD simulation
scheme = RK4           # 4th-order Runge-Kutta numerical scheme
start_simulation = 2024-01-01 23:00:00
end_simulation = 2024-02-10 23:00:00
write_frequency = 3600 # Output written every 1 hour

[Diffusion]
Kh = 1.0
Kz = 0.001

[Files]
cmems_dir = /mnt/d/CMEMS_Indo/
era5_dir = /mnt/d/ERA5_native/
output_dir = /mnt/d/output/
source_csv = forward_sources.csv

[Release]
start_date_release = 2024-01-01 22:00:00
```

```
end_date_release = 2024-02-10 22:00:00
release_rate_per_day = 2
diameter = 1000

[MarineDebris]
type = macro_plastic
leeway_macro_plastic = 0.03
beaching_probability = 0.05
```

4.3 Example Configuration: Backward Origin Tracking

For backward tracking simulations (e.g., backtracking the source of debris intercepted in the Banda Sea or Makassar Strait), set the time step 'dt' to a negative value (e.g., dt = -3600), with start_simulation representing the observation date and end_simulation pointing to an earlier date in the past.

4.4 Example Configuration: 3D Microplastic Transport with Biofouling

The configuration file namelist_microplastic_forward_Rk4_2.5D.txt parameterizes vertical settling dynamics:

```
[Simulation]
mode = microplastic
dimension = 3DUV       # Use 3DUVW if vertical ocean velocity 'w' is available
dt = 3600
scheme = RK4

[Microplastic]
polymer_type = PET     # Polyethylene Terephthalate (~ 1380 kg/m3)
diameter_mm = 0.1      # 100 micron particle diameter
shape = sphere
ws_type = static       # Options: static or dynamic
ws_static = 0.0001     # Static settling velocity in m/s
biofilm_growth_rate = 0.001
biofilm_density = 1400.0
```

CHAPTER 5: INPUT DATASET SPECIFICATIONS

5.1 Hydrodynamic Ocean Current Fields (CMEMS Reanalysis / Forecast)

LTM assimilates 3D ocean velocity data from Copernicus Marine Environment Monitoring Service (CMEMS). NetCDF files stored in cmems_dir must comply with the following standards:

- **Horizontal Grid:** Monotonically increasing regular longitude and latitude 1D/2D arrays.
- **Vertical Coordinate:** Standard depth levels in meters, positive downwards (0 meters at the surface).
- **Velocity Variables:** Zonal velocity 'uo' (m/s) and meridional velocity 'vo' (m/s). If 3DUVW mode is active, vertical velocity 'wo' (m/s) must be provided.
- **Time Coordinate:** CF-compliant time dimension, e.g., 'hours since 1950-01-01 00:00:00'.

5.2 Surface Atmospheric Wind Fields (ERA5 Reanalysis 10m Wind)

Atmospheric wind forcing is obtained from the ECMWF ERA5 Reanalysis dataset at 10 meters above the sea surface. NetCDF files in era5_dir must supply u10 (10m zonal wind component) and v10 (10m meridional wind component).

5.3 Seeding Source CSV File Specifications

Particle seeding stations are defined in CSV format without header rows, containing 4 required fields: station_name, longitude (°E), latitude (°N), and release depth (m). Representative source locations provided with the model include:

Source Station	Longitude (°E)	Latitude (°N)	Release Depth (m)
jakarta (forward_sources.csv)	106.922588	-5.929110	0 (Surface)
North_pacific (forward_sources.csv)	127.229219	7.785793	0 (Surface)
South_pacific (forward_sources.csv)	132.817523	-0.190275	0 (Surface)
banda (backward_sources.csv)	130.975562	-5.724162	1000 (Deep Sea)
Sulawesi_sea (backward_sources.csv)	122.007029	3.492565	500 (Water Column)
SCS / Natuna Sea (backward_sources.csv)	108.447786	5.934744	500 (Water Column)
South_java (backward_sources.csv)	106.173698	-8.881424	500 (Water Column)
Java sea (backward_sources.csv)	110.504903	-4.920708	5 (Near Surface)

Ciliwung River (river_indo_marine_debris.csv)	106.800000	-6.080000	0 (River Mouth)
Citarum River (river_indo_marine_debris.csv)	106.960000	-5.920000	0 (River Mouth)
Mahakam River (river_indo_marine_debris.csv)	117.600000	-0.320000	0 (River Mouth)
Barito River (river_indo_marine_debris.csv)	114.480000	-3.520000	0 (River Mouth)

Note: The included dataset river_indo_marine_debris.csv catalogs over 50 major river discharge points across all main Indonesian islands (Sumatra, Java, Kalimantan, Sulawesi, Bali-Nusa Tenggara, Maluku, and Papua), facilitating comprehensive nationwide marine debris transport studies.

CHAPTER 6: MATHEMATICAL FORMULATIONS & PARTICLE DYNAMICS

All physical processes and governing numerical equations in RCOTrack / LTM are formatted below according to the native Microsoft Office OpenXML Math (OMML) standard.

6.1 Lagrangian Kinematic Transport Equation

The spatial trajectory of particle i at time t is described by the 3D position vector r_i(t) = (λ_i, ϕ_i, z_i), where λ denotes longitude, ϕ denotes latitude, and z represents depth (positive downwards). The temporal evolution of particle position is governed by the advection-diffusion differential equation:

$$\frac{d\mathbf{r}}{dt} = \mathbf{U}_{total}(\mathbf{r}, t) + \mathbf{U}_{diffusion}(\mathbf{r}, t)$$

Equation 6.1: Governing 3D Lagrangian Kinematic Advection-Diffusion Equation

On the spherical terrestrial surface, the horizontal velocity components (Ux, Uy) are converted to angular displacements via:

$$\frac{d\lambda}{dt} = \frac{U_x}{R_E \cos(\varphi)}, \quad \frac{d\varphi}{dt} = \frac{U_y}{R_E}$$

Equation 6.2: Transformation from Metric Velocity to Spherical Angular Coordinates

6.2 Effective Surface Velocity & Wind Leeway Effect

In 'marine_debris' mode (2.5D), particles in the ocean surface layer experience direct atmospheric drag (wind leeway drift) in conjunction with surface ocean currents (u_ocean, v_ocean):

$$U_x = u_{ocean} + L_w \cdot U_{10}, \quad U_y = v_{ocean} + L_w \cdot V_{10}$$

Equation 6.3: Total Horizontal Velocity with Empirical Wind Leeway Factor

Where L_w is the dimensionless leeway coefficient configured in namelist.txt:

- General macroplastic (macro_plastic): L_w = 0.03 (3% of 10m wind velocity)
- Wood debris (wood): L_w = 0.02 (2%)
- Expanded polystyrene / Styrofoam (styrofoam): L_w = 0.05 (5%)
- Search-and-rescue target (human): L_w = 0.04 (4%)

6.3 Numerical Integration Schemes

LTM implements three temporal integration schemes to advance particle positions from t to t + Δt:

Scheme 1: 1. 1st-Order Explicit Euler Scheme:

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{U}(\mathbf{r}(t), t) \cdot \Delta t$$

Equation 6.4: Explicit First-Order Euler Scheme

Scheme 2: 2. 2nd-Order Runge-Kutta (RK2 / Heun) Scheme:

$$\mathbf{r}_{pred} = \mathbf{r}(t) + \mathbf{U}(\mathbf{r}(t), t) \cdot \Delta t$$

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \frac{1}{2}\left[\mathbf{U}(\mathbf{r}(t), t) + \mathbf{U}\left(\mathbf{r}_{pred}, t + \Delta t\right)\right] \cdot \Delta t$$

Equation 6.5: Second-Order Runge-Kutta (Predictor-Corrector) Scheme

Scheme 3: 3. 4th-Order Classical Runge-Kutta (RK4) Scheme:

$$\mathbf{k}_1 = \mathbf{U}(\mathbf{r}(t), t), \quad \mathbf{k}_2 = \mathbf{U}\left(\mathbf{r} + \frac{\Delta t}{2}\mathbf{k}_1, t + \frac{\Delta t}{2}\right)$$

$$\mathbf{k}_3 = \mathbf{U}\left(\mathbf{r} + \frac{\Delta t}{2}\mathbf{k}_2, t + \frac{\Delta t}{2}\right), \quad \mathbf{k}_4 = \mathbf{U}(\mathbf{r} + \Delta t \cdot \mathbf{k}_3, t + \Delta t)$$

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \frac{\Delta t}{6}[\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4]$$

Equation 6.6: Classical 4th-Order Runge-Kutta (RK4) Scheme

6.4 Turbulent Random Walk Diffusion

Sub-grid scale turbulent dispersion is modeled as a stochastic Random Walk process. Position increments resulting from horizontal (Kh) and vertical (Kz) turbulent diffusion are given by:

$$\Delta r_h = R_{n,h}\sqrt{2 \cdot K_h \cdot |\Delta t|}, \quad \Delta z_{diff} = R_{n,z}\sqrt{2 \cdot K_z \cdot |\Delta t|}$$

Equation 6.7: Stochastic Horizontal and Vertical Random Walk Dispersions

Where R_n,h and R_n,z are independent standard normal random variables drawn from N(0, 1) at each time step.

6.5 3D Microplastic Settling & Biofouling Dynamics

In 'microplastic' mode, particles undergo full 3D displacement influenced by vertical ocean advection (w), vertical turbulent diffusion (Kz), and settling/buoyancy velocity (ws):

$$\frac{dz}{dt} = w_{ocean} + w_s + \left(\frac{dz}{dt}\right)_{diffusion}$$

Equation 6.8: Vertical Particle Motion Governing Equation

When dynamic settling is activated, microbial and algal biofilm growth increases particle mass and reduces buoyancy. The effective particle density is governed by the volume-weighted composite formulation (Kooi et al., 2017):

$$\rho_{eff}(t) = \frac{\rho_p V_p + \rho_{bf} V_{bf}(t)}{V_p + V_{bf}(t)}$$

Equation 6.9: Composite Biofouled Microplastic Effective Density

The terminal settling velocity (ws) for spherical microplastics is calculated via Stokes' Law formulation:

$$w_s = \frac{2}{9}\frac{(\rho_{eff} - \rho_{sw}) \cdot g \cdot {r_{eff}}^2}{\mu_{sw}}$$

Equation 6.10: Terminal Settling Velocity Formulation

Where ρ_sw is seawater density (~1025 kg/m³), g is gravitational acceleration (9.81 m/s²), r_eff is effective biofouled radius, and μ_sw is dynamic seawater viscosity (~0.001 kg/(m·s)).

6.6 Shoreline Beaching & Stranding Mechanics

Particle-land boundary interactions are governed by bathymetry depths (io.bathy_depth) and coastal mask proximity. When a particle encounters shallow coastal waters (< 1 meter) or crosses land boundaries, it is evaluated against beaching_probability (typical: 0.05). If stranded, its status flag is updated to '0' (stranded), locking its position at the shoreline and removing it from active hydrodynamic advection for the remainder of the simulation.

CHAPTER 7: SIMULATION EXECUTION & OUTPUT MANAGEMENT

7.1 Step-by-Step Linux Execution Workflow

Follow this standard procedure to initiate an LTM simulation in Linux Bash:

- **Step 1:** Verify that all CMEMS current and ERA5 wind NetCDF files are populated in the specified directories.
- **Step 2:** Configure namelist.txt (or copy one of the pre-configured namelist_*.txt templates).
- **Step 3:** Confirm that the release location CSV file contains the desired station coordinates.
- **Step 4:** Execute the main driver script from the active Conda environment in Linux terminal:

```
# Navigate to root model directory (Linux POSIX path)
cd /path/to/RCOTrack

# Ensure the lagrangian environment is active
conda activate lagrangian

# Run simulation
python main.py
```

7.2 Console Logging & Progress Monitoring

During execution, main.py displays formatted diagnostic headers and an interactive tqdm progress bar:

```
--- [PROCESS] INITIALIZING INPUT DATA ---
--- [INFO] MODE: marine_debris | DIRECTION: FORWARD | SCHEME: RK4 ---
Particles evolve:  45%|████████████              | 432/961 [04:12<05:08,
1.72step/s]
...
--- [PROCESS] SIMULATION COMPLETED SUCCESSFULLY ---
--- [INFO] AUTHOR: Dr Dwiyoga Nugroho, RCO-BRIn 2026 ---
```

7.3 Computational Performance & Memory Optimization

To ensure uninterrupted multi-month simulations without memory exhaustion, LTM implements dual memory safeguards:

1. Periodic Garbage Collection: Every 86,400 elapsed simulation seconds (1 day), memory garbage collection (gc.collect()) is triggered, and previous step velocity tensors are explicitly freed (del u, v, w, uw, vw).

2. Zero-Copy C-Contiguous Memory Arrays: The update_3d_safe module converts coordinates into continuous C-order arrays (np.float64 / np.int32) prior to invoking compiled Numba kernels, maximizing CPU L1/L2 cache utilization.

7.4 Output NetCDF-4 File Schema

Trajectory results are written to output_dir with standardized naming conventions, e.g.: test_run_marine_debris_forward_20240101_20240210.nc. The internal NetCDF variable schema includes:

Variable Name	Dimensions / Type	Description & Physical Units
time	(time) / float64	Simulation step timestamp (seconds since 1970-01-01 00:00:00).
particle_id	(obs) / int32	Globally unique numeric identifier assigned to each released particle.
lon	(obs) / float64	Particle longitudinal position in decimal degrees (°E).
lat	(obs) / float64	Particle latitudinal position in decimal degrees (°N).
z	(obs) / float64	Particle depth coordinate in meters (0 = surface).
status	(obs) / int32	Particle status flag: 1 = actively drifting; 0 = beached / stranded ashore.

CHAPTER 8: POST-PROCESSING & SPATIAL VISUALIZATION

8.1 Marine Debris Trajectory Mapping (plot_marine_debris.py)

The plot_marine_debris.py utility generates publication-quality cartographic figures using Cartopy and Matplotlib. It automatically recognizes simulation direction from NetCDF global metadata:

• Forward Tracking: Initial release locations are rendered as RED circles, trajectory pathways as thin grey lines, and final positions as BLACK dots.

• Backward Tracking: Initial observation points are rendered as BLACK circles, reverse trajectories as thin grey lines, and inferred origins as RED dots.

• Stranded Particles: Particles reaching the coast (status = 0) are highlighted with BLUE crosses (x).

```
# Configuration within plot_marine_debris.py
NC_FILE = 'output/test_run_marine_debris_backward_20240210_20240101.nc'
OUTPUT_IMAGE = 'marine_debris_backward_map.jpg'
PRODUCE_ANIMATION = False

# Generate map figure in Linux terminal
python plot_marine_debris.py
```

8.2 3D Microplastic Visualization & Animation (plot_microplastic.py)

The plot_microplastic.py script visualizes vertical dispersion throughout the water column:

• Uses Matplotlib LineCollection with the 'jet' colormap to display particle depth (meters).

• Features an inverted vertical colorbar with surface (0 m) at the top, consistent with oceanographic standards.

• Produces interactive multi-frame GIF animations (PRODUCE_ANIMATION = True) utilizing PillowWriter with customizable particle trail limits.

```
# Configure animated trajectory generation in plot_microplastic.py:
PRODUCE_ANIMATION = True
FRAME_SKIP = 2
TRAIL_LIMIT = 20

python plot_microplastic.py
# Output file: test_run_microplastic_3DUV_forward_20240101_20240210.gif
```

CHAPTER 9: TROUBLESHOOTING & TECHNICAL REFERENCES

9.1 Numerical Stability: The Courant-Friedrichs-Lewy (CFL) Condition

When particles move through high-velocity currents (such as the narrow straits of Lombok, Alas, or Sape), an excessively large time step (dt) can cause particles to overshoot grid cells and artificially cross land barriers. The time step must satisfy the CFL stability criterion:

$$C = \frac{|\mathbf{U}| \cdot \Delta t}{\Delta x} \leq 1 \quad \Longrightarrow \quad \Delta t \leq \frac{\Delta x}{|\mathbf{U}|_{max}}$$

Equation 9.1: Courant-Friedrichs-Lewy (CFL) Numerical Stability Constraint

9.2 Common Troubleshooting Scenarios

Operating System Error: Attempting to run natively in Windows CMD / PowerShell

Cause: Windows cmd.exe or PowerShell does not support bash scripts, uses non-POSIX backslash '\' paths, and encounters DLL import mismatches with compiled Numba/OpenMP and NetCDF C-libraries.
Resolution: MANDATORY LINUX REQUIREMENT: You MUST run the program inside Linux (Ubuntu, Rocky Linux, CentOS) or inside WSL2 (Windows Subsystem for Linux) using bash.

Repository Cloning / Download Issues

Cause: Incomplete clone from GitHub or missing Git package on the compute server. Resolution: Install git via your package manager ('sudo apt install git' or 'sudo dnf install git') and clone directly from https://github.com/dwygngrh/RCOTrack.git, or use wget to fetch the ZIP archive.

FileNotFoundError: CMEMS or ERA5 Data Missing.

Cause: The path configured in namelist.txt (cmems_dir or era5_dir) is incorrect or daily NetCDF files are incomplete. Resolution: Check POSIX directory paths, ensure trailing slashes '/' are present (e.g. /mnt/d/CMEMS_Indo/), and verify that the date range matches the simulation span.

OutOfMemoryError / RAM Exhaustion During Simulation

Cause: Excessive particle release counts or loading full uncropped regional ocean domains into memory simultaneously. Resolution: Reduce release_rate_per_day, increase write_frequency (e.g., from 1800s to 3600s), and verify that io.get_step_data applies geographic bounding box cropping.

Particles Jumping Across Landmasses (CFL Violation)

Cause: The numerical time step (dt) is too large relative to local flow velocities in high-speed tidal straits (e.g., Lombok or Alas Strait). Resolution: Decrease dt from 3600s (1 hr) to 1800s (30 min) or 900s (15 min), and ensure the numerical scheme is set to 'RK4'.

Cartopy Download Error in Offline HPC Compute Nodes

Cause: HPC compute nodes without direct outbound internet access cannot automatically download Natural Earth coastline shapefiles. Resolution: Pre-download Natural Earth shapefiles and place them in the local user cache directory at ~/.local/share/cartopy/shapefiles/natural_earth/.

9.3 Scientific References

- **Biofouling Dynamics:** Kooi, M., et al. (2017). The effect of particle properties on the depth profile of buoyant plastics in the ocean. Scientific Reports, 7(1), 13371.
- **Wind Leeway Modeling:** Breivik, Ø., et al. (2011). Wind-induced drift of objects at sea: The leeway field method. Applied Ocean Research, 33(2), 100-110.
- **Model Architecture:** Nugroho, D., et al. (2026). RCOTrack: High-Performance 3D Lagrangian Particle Trajectory Modeling Engine for Indonesian and Regional Waters. Technical Report, Research Center for Oceanography - BRIN, Jakarta.